

This is the non-calculator section!

Period 5For 1 and 2, solve for x , leaving answers in exact form with powers and logs. [3 pts. each]

-1 ✓ 1. $2 \ln(\sqrt{x}) - \ln(1-x) = 2$

$$\ln(\sqrt{x})^2 - \ln(1-x) = 2$$

$$\ln x - \ln(1-x) = 2$$

$$\ln\left(\frac{x}{1-x}\right) = 2$$

$$\frac{x}{1-x} = e^2$$

$$x = e^2(1-x) = e^2 - e^2x$$

$$x - e^2x = e^2$$

$$x(1-e^2) = e^2$$

$$x = \frac{e^2}{1-e^2}$$

✓ 2. $\frac{10-3^x}{4} + 5 = -20$

$$\frac{10-3^x}{4} = -25$$

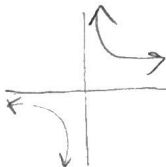
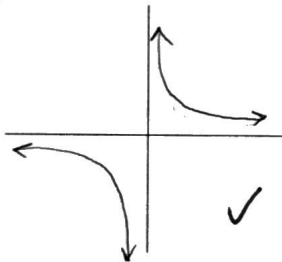
$$10-3^x = -100$$

$$3^x = 110$$

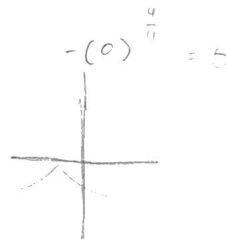
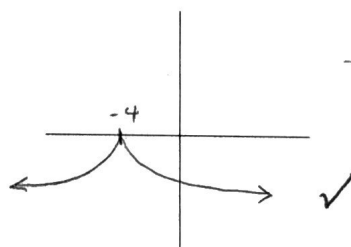
$$x = \log_3 110$$
 ✓

Draw a rough sketch of the following 2 power functions. We are looking for shapes and domain and ranges so your graphs don't have to be super exact. [3 pts. each]

✓ 3. $f(x) = (x)^{-\frac{5}{9}}$



✓ 4. $f(x) = -(x+4)^{\frac{4}{11}}$

-1 ✓ 5. Write an equation of a logistic curve in the form $f(x) = \frac{M}{1+Be^{-kx}}$ that has a point of inflection at (40, 2000). [2 pts]

$$\frac{2,000}{2} = 1,000$$

$$\frac{M}{1+e^{-kx}}$$

$$\rightarrow M = 2,000 \times 2 = 4,000$$

$$\rightarrow \frac{4,000}{1+e^{-kx}}$$

$$\rightarrow \frac{4,000}{1+e^{-x+40}}$$

$$\left. \begin{aligned} 1+e^{-x+40} &= 2 \\ e^{-x+40} &= 1 \\ -x+40 &= 0 \\ x &= 40 \end{aligned} \right\} \checkmark$$

6. Given the following logistic function: $f(x) = \frac{60}{1+5e^{-0.2x}}$

a) Find the coordinates of the y-intercept and determine the carrying capacity. [2]

y intercept $(0, 10)$ ✓ Carrying capacity 60 ✓

$$\frac{60}{1+5e^0}$$

$$0 = -5 \ln \left(\frac{60-x}{5x} \right)$$

$$0 = \ln \left(\frac{60-x}{5x} \right)$$

$$1 = \frac{60-x}{5x} \quad 5x = 60-x$$

$$6x = 60$$

$$x = 10$$

Find the inverse function $f^{-1}(x)$ by switching x and y and solving for y. [4 pts]

$$x = \frac{60}{1+5e^{-0.2y}}$$

$$x(1+5e^{-0.2y}) = 60$$

$$1+5e^{-0.2y} = \frac{60}{x}$$

$$5e^{-0.2y} = \frac{60}{x} - 1$$

$$e^{-0.2y} = \frac{\frac{60}{x} - 1}{5} = \frac{60-x}{5x}$$

$$-0.2y = \ln \left(\frac{60-x}{5x} \right)$$

$$y = -5 \ln \left(\frac{60-x}{5x} \right) \quad \checkmark$$

$$-0.2 = -\frac{2}{10} = -\frac{1}{5}$$

b) Write equations of all asymptotes for $f^{-1}(x)$. [1 pt]

$x=0$ $x=60$ ✓

$$\frac{60-x}{5x} > 0$$

$$60-x > 0$$

$$60 > x$$

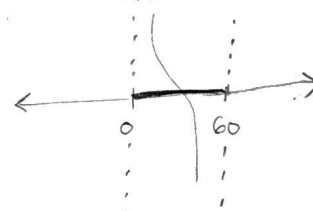
$$x < 60$$

$$x \neq 0$$

$$x > 0$$

$$5x \neq 0$$

$$x \neq 0$$



$$\frac{60-61}{0}$$

$$60 - (-1) = 61$$

Continue on to the calculator section. Before taking out your calculator you must turn in this non-calculator section.

Name again: My problems are compounding continuously! cute chang

Period 5

This is the calculator page! Turn in your other sheet before taking out your calculator

$$FV = C \left[\frac{(1+i)^n - 1}{i} \right] \quad PV = C \left[\frac{1 - (1+i)^{-n}}{i} \right]$$

7. You have two different investment opportunities that you are trying to choose between.

Option 1: Invest \$3000 in a CD (certificate of deposit) that grows at a 4% annual interest rate (compounded monthly). The CD requires you to leave it "untouched" for 5 years.

Option 2: Invest \$60 every month for 5 years at an annual interest rate of 3%.

- a) Both options involve investing a total of \$3000 over 5 years. Find the Future Value for both options. [4 pts.]

$$A = 3,000 \left(1 + \frac{0.04}{12} \right)^{12 \cdot 5}$$

$$A = 3462.99$$

$$FV = 60 \left(\frac{\left(1 + \frac{0.03}{12} \right)^{5 \cdot 12} - 1}{\frac{0.03}{12}} \right)$$

$$FV = 3878.8$$

FV Option 1 \$3,662.99 ✓

FV Option 2 \$3,878.80 ✓

- b) How would the annual interest rate have to change in "Option 1" so that the two investments would have the same Future Value? Round your interest rate to the nearest hundredth place. [3 pts]

$$3,000 \left(1 + \frac{r}{12} \right)^{12 \cdot 5} = 3878.8$$

$$r = \boxed{5.15\%}$$

it would have to increase by 0.15%

Oh! sec. Sure!

- c) Give a financial reason that you might actually choose the investment with the SMALLER Future Value. [0 points. Obviously don't spend too much time thinking about this one!]

you're too lazy to put \$60 in every month and just wanna put the whole

Turn over for the last problem!

\$3,000 in at once

8. You have your eye on a \$35,000 used Dodge Charger car, and will pay for it by paying partly with cash (putting money down) and partly with financing (taking a loan). The car dealership is offering 5% annual interest rate on a 3-year loan. You know that you can only afford \$500 monthly payments. How much cash would you have to put down to stay within your monthly budget? [5 pts].

PV question

$$PV = 500 \left(\frac{1 - \left(1 + \frac{0.05}{12}\right)^{-12 \cdot 3}}{\frac{0.05}{12}} \right) = 16682.9$$

$$35,000 - PV = 35,000 - 16,682.9 = \boxed{\$18,317.1}$$

Do 2 decimal places for money.